

Geometry & Coordinate Geometry



Distance and midpoint

- 1 Find the distance between $(1, 2)$ and $(4, 6)$.
 - 2 Find the midpoint of $(2, -3)$ and $(8, 5)$.
 - 3 Find the distance between $(-2, 1)$ and $(3, -11)$.
 - 4 Find the midpoint of $(-4, 7)$ and $(6, -3)$.
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Gradient (slope)

- 5 Find the gradient of the line through $(1, 2)$ and $(5, 10)$.
 - 6 Find the gradient of the line through $(-3, 4)$ and $(2, -6)$.
 - 7 Find the gradient of the line through $(0, 0)$ and $(6, 9)$.
 - 8 State whether the line through $(1, 1)$ and $(4, 1)$ is horizontal, vertical, or neither.
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Equations of lines

- 9 Find the equation of the line with gradient 2 passing through $(3, 5)$, in the form $y = mx + c$.
 - 10 Find the equation of the line through $(0, 4)$ and $(2, 10)$.
 - 11 Find the x - and y -intercepts of the line $2x + 3y = 12$.
 - 12 Write the equation of the line parallel to $y = 4x - 1$ passing through $(2, 9)$.
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Parallel and perpendicular lines

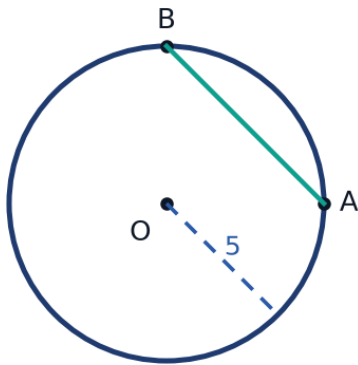
- 13 State the gradient of a line perpendicular to a line with gradient 5.
- 14 Determine whether the lines $y = 2x + 3$ and $y = 2x - 7$ are parallel, perpendicular, or neither.
- 15 Find the equation of the line perpendicular to $y = -\frac{1}{3}x + 2$, passing through $(3, 4)$.

The equation of a circle

- 16 Write the equation of a circle centered at the origin with radius 6.
- 17 Write the equation of a circle centered at $(2, -3)$ with radius 5.
- 18 Find the radius and center of the circle $(x - 1)^2 + (y + 4)^2 = 49$.
- 19 Find the equation of the circle centered at the origin that passes through $(3, 4)$.

Circles: chords and arcs

- 20 Points A and B lie on a circle of radius 5 centered at O , with central angle $\angle AOB = 90^\circ$. Find the length of chord AB , using $AB = 2r \sin\left(\frac{\theta}{2}\right)$.



- 21 A chord of a circle with radius 10 cm is 12 cm long. Find the distance from the center to the chord.
- 22 Find the circumference of a circle with radius 9 cm, in terms of π .

Polygon angle sums

- 23 Find the sum of the interior angles of a hexagon (6 sides).
- 24 Find the measure of each interior angle of a regular pentagon.
- 25 A polygon has an interior angle sum of 1440° . Find the number of sides.

Perimeter and area

- 26 A rectangle has length 12 and width 7. Find its perimeter and area.
- 27 A right triangle has legs 6 and 8. Find its area and hypotenuse.
- 28 A regular hexagon has side length 4, with area $A = \frac{3\sqrt{3}}{2}s^2$. Find its area in exact form.

Coordinate geometry: identifying shapes

- 29 Show that the triangle with vertices $A(0, 0)$, $B(4, 0)$, $C(4, 3)$ is a right triangle.
- 30 Determine whether the points $A(-2, 1)$, $B(2, 4)$, $C(5, 0)$ form a right triangle, using the squared side lengths.
- 31 Determine whether the triangle with vertices $A(1, 2)$, $B(5, 2)$, $C(3, 6)$ is isosceles.

The perpendicular bisector of a segment

- 32 Find the equation of the perpendicular bisector of the segment joining $(2, 3)$ and $(8, 7)$.
- 33 Find the equation of the perpendicular bisector of the segment joining $(0, 0)$ and $(6, 8)$.
- 34 A line segment has endpoints $(1, 5)$ and $(7, 5)$. Find the equation of its perpendicular bisector.

Word problems: geometry on a coordinate grid

- 35 This question has two parts. A city park is designed on a coordinate grid (in meters), with two gates at $A(0, 0)$ and $B(80, 60)$.
- a Find the straight-line distance between the two gates.
- b A fountain is placed at the midpoint between the two gates. Find its coordinates.

- 36** This question has two parts. A circular irrigation sprinkler is centered at $(5, 5)$ on a farm's coordinate grid (in meters) and has a radius of 12 meters.
- a** Write the equation of the circle covered by the sprinkler.
 - b** Determine whether a tree at $(14, 9)$ lies inside, on, or outside the watered region.